

Calculating points for decathlon and heptathlon

The 2001 IAAF points tables use the following formulae:

- Points = $\text{INT}(A(B - P)^C)$ for track events (faster time produces a higher score)
- Points = $\text{INT}(A(P - B)^C)$ for field events (greater distance or height produces a higher score)

A, **B** and **C** are parameters that vary by discipline, as shown in the table below, while **P** is the performance by the athlete, measured in seconds (running), metres (throwing), or centimetres (jumping). INT means that the points are always rounded down e.g. 599.9999 = 599

Decathlon table

Event	A	B	C
100 m	25.4347	18	1.81
Long jump	0.14354	220	1.4
Shot put	51.39	1.5	1.05
High jump	0.8465	75	1.42
400 m	1.53775	82	1.81
110 m hurdles	5.74352	28.5	1.92
Discus throw	12.91	4	1.1
Pole vault	0.2797	100	1.35
Javelin throw	10.14	7	1.08
1500 m	0.03768	480	1.85

Heptathlon table

Event	A	B	C
200 metres	4.99087	42.5	1.81
800 metres	0.11193	254	1.88
100 metres hurdles	9.23076	26.7	1.835
High jump	1.84523	75	1.348
Long jump	0.188807	210	1.41
Shot put	56.0211	1.5	1.05
Javelin throw	15.9803	3.8	1.04

Example: Decathlon 100 m run in 11.00 seconds

$$\begin{aligned}\text{Points} &= \text{INT}(A(B - P)^C) = \text{INT}(25.4347(18-11)^{1.81}) = \text{INT}(25.4347 \cdot 7^{1.81}) = \\ &= \text{INT}(25.4347 \cdot 33.8553612636) = \text{INT}(861.100957131) = 861\end{aligned}$$

Points = 861

Example: Heptathlon shot put for 16.5 meters

$$\begin{aligned}\text{Points} &= \text{INT}(A(P - B)^C) = \text{INT}(56.0211(16.5-1.5)^{1.05}) = \text{INT}(56.0211 \cdot (15)^{1.05}) = \\ &= \text{INT}(56.0211 \cdot 17.1749634697) = \text{INT}(962.160346032) = 962\end{aligned}$$

Points = 962